Theorem
If X and Y are Hausdorff topological spaces, then so is $X \times Y$.

Proof: Let (x_1, y_1) and (x_2, y_2) be distinct points in $X \times Y$. Then $x_1 \neq x_2$ or $y_1 \neq y_2$.

Suppose $x_1 \neq x_2$. Since X is Hausdorff, there are disjoint sets U_1 and U_2 open in X such that $x_1 \in U_1$ and $x_2 \in U_2$.

Then $U_1 \times Y$ and $U_2 \times Y$ are open neighborhoods of (x_1, y_1) and (x_2, y_2) respectively, and

$$(U_1 \times Y) \cap (U_2 \times Y) = \varnothing.$$

A similar argument holds if $y_1 \neq y_2$. Therefore, $X \times Y$ is Hausdorff. ∎

Theorem
If X and Y are nonempty topological spaces, and $X \times Y$ is Hausdorff, then X is also Hausdorff.

Proof: Let x_1 and x_2 be distinct points in X. Because Y is nonempty, choose another point $y \in Y$.

Then (x_1, y) and (x_2, y) are distinct points in $X \times Y$. Since $X \times Y$ is Hausdorff, there are disjoint open sets $W_1, W_2 \subseteq X \times Y$ such that $(x_1, y) \in W_1$ and $(x_2, y) \in W_2$.

Moreover, because W_1 and W_2 are open,

$$(x_1, y) \in U_1 \times V_1 \subseteq W_1$$
$$(x_2, y) \in U_2 \times V_2 \subseteq W_2$$

for some basic open neighborhoods $U_1 \times V_1$ and $U_2 \times V_2$.

Then $x_1 \in U_1$ and $x_2 \in U_2$. We now show U_1 and U_2 are disjoint. Assume for contradiction that $u \in U_1 \cap U_2$. Then

$$(u, y) \in U_1 \times V_1 \subseteq W_1$$
$$(u, y) \in U_2 \times V_2 \subseteq W_2$$

which would contradict $W_1 \cap W_2 = \varnothing$.

So the open sets U_1 and U_2 separate the points x_1 and x_2. Therefore, X is a Hausdorff topological space. ∎

Theorem

If X and Y are path-connected topological spaces, then $X \times Y$ is also path-connected.

Proof: Let (x_0, y_0) and (x_1, y_1) be points in $X \times Y$. Then x_0 and x_1 are connected by a path, and so are y_0 and y_1. Let f and g be these paths respectively:

$$f : [0,1] \to X \quad f(0) = x_0 \text{ and } f(1) = x_1$$
$$g : [0,1] \to Y \quad g(0) = y_0 \text{ and } g(1) = y_1$$

Let F be a path between (x_0, y_0) and (x_1, y_1) defined as

$$F : [0,1] \to X \times Y,$$
$$F(t) = (f(t), g(t)).$$

Then $F(0) = (x_0, y_0)$ and $F(1) = (x_1, y_1)$.

A theorem in the text states that if all projections of a function f to a product are continuous, then f is also continuous.

In this case, $\pi_X \circ F = f$ and $\pi_Y \circ F = g$ are continuous because they are paths, so F is also continuous. However, we can prove the continuity of F directly.

Suppose $W \subseteq X \times Y$ is open. Let $t \in F^{-1}(W)$. Then $F(t) \in W$. Hence

$$(f(t), g(t)) \in U \times V \subseteq W$$

for some sets U open in X and V open in Y.

Thus $f(t) \in U$ and $g(t) \in V$, meaning

$$t \in f^{-1}(U) \cap g^{-1}(V) \subseteq F^{-1}(W)$$

which by the continuity of f and g is an intersection of open sets. We have thus proven $F^{-1}(W)$ is open one point at a time.

Therefore, F is a path between (x_0, y_0) and (x_1, y_1). Since these points were arbitrary, it follows that $X \times Y$ is path-connected. ■